

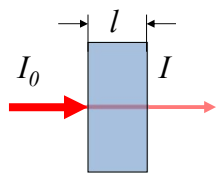
# Introduction to Biophysics II: Quantum Effects in Biology

## Lecture 05: Measuring light. Experimental techniques

- Steady state spectroscopy:
  - Absorption UV-VIS spectrometer, Fourier spectrometer
  - Hole-burning spectroscopy
  - Photoacoustic spectroscopy
  - Emission spectroscopy: fluorimeter
  - Circular dichroism spectroscopy
- Fluorescence: scale conversions
- Measuring biological samples: challenges
- Time-resolved (transient) spectroscopy:
  - Pump-probe spectroscopy,
  - Time-correlated photon counting.
  - Up-conversion

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## Absorption: Beer-Lambert Law



Light is absorbed in the sample, and intensity lowers exponentially with optical path:

$$\frac{dI}{dx} = -\epsilon'IC$$

$$I = I_0 \exp(-\epsilon'Cl) = I_0 10^{-\epsilon Cl} \equiv I_0 10^{-A}$$

$$A = \log \frac{I_0}{I} = \epsilon cl$$

Transmission:

$$T = I/I_0$$

$$T = 10^{-A}$$

$I_0$  - incident light intensity

$I$  - transmitted light intensity-

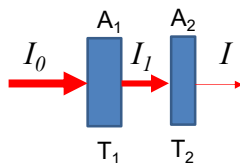
$l$  - sample thickness (cm)

$c$  - molar concentration of a sample (mol/L)

$\epsilon$  - molar extinction coefficient, (L mol<sup>-1</sup>cm<sup>-1</sup>)

(absorbance of 1 cm thick sample at 1 mol/L)

Why log? → additivity of absorption:  $I = T_2 I_1 = T_1 T_2 I_0 = T I_0$



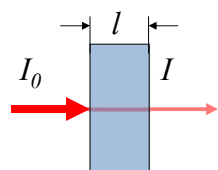
Total transmission:  $T = T_1 T_2$

Total absorption:  $A = A_1 + A_2$

*Absorption is additive, proportional to concentration*

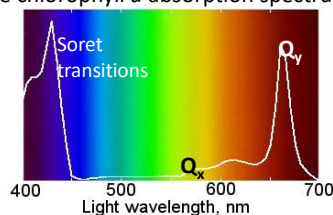
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## Absorption: Beer-Lambert Law



$$A = \log \frac{I_0}{I} = \epsilon c l$$

The chlorophyll *a* absorption spectrum



**The extinction coefficient**  $\epsilon$  relates to the probability of absorbing a photon, and it depends on the wavelength of a photon.

$\epsilon$  units  $\rightarrow$  ( $\text{L mol}^{-1}\text{cm}^{-1}$ )  $\rightarrow$  area/mol  $\rightarrow$  area per molecule. i.e.  $\sim$ the effective area of a molecule over which it absorbs a photon

It thus relates to the transition probability (transition dipole) derived earlier:

$$R_{a \rightarrow b} = \frac{\pi}{3\epsilon_0 \hbar^2} |\vec{\mu}|^2 \rho(\omega_0) \quad \vec{\mu} \equiv e \langle \psi_b | \vec{r} | \psi_a \rangle$$

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## Absorption: oscillator strength

$$R_{a \rightarrow b} = \frac{\pi}{3\epsilon_0 \hbar^2} |\vec{\mu}|^2 \rho(\omega_0) \quad \vec{\mu} \equiv e \langle \psi_b | \vec{r} | \psi_a \rangle$$

**Oscillator strength,  $f$** , is a dimensionless quantity that expresses the probability of absorption or emission of electromagnetic radiation

Definition: The oscillator strength of a given transition is defined in terms of the transition probability of a three-dimensional isotropic harmonic oscillator having mass and charge equal to the electron's mass and charge, and vibrational frequency equal to the frequency of the electronic transition in question

$\rightarrow$  Our expression for  $R_{a \rightarrow b}$  should be divided by a similar expression calculated for the transition of a 3D oscillator for the electron (i.e. calculate  $\mu$ ):

$$f = \frac{R_{a \rightarrow b}}{R_{a \rightarrow b}^{\text{harm}}} = \frac{2}{3} \frac{m_e}{\hbar^2 e^2} \Delta E |\vec{\mu}|^2$$

*Details: Principles and Applications of Quantum Chemistry, 2016, Pages 291-337*

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## Oscillator strength: the reference

Oscillator strength  $f$  is measured with respect to the *classical* light absorption probability by a one-electron atom.

1. Light EM wave causes the electron to oscillate under periodic force  $F = eE$ , where  $E$  is the oscillating electric field and  $e$  is the electron charge.
2. As an electron is accelerated, it emits EM radiation at the same frequency. Which is equivalent to the absorption of incident light (energy conservation)
3. The rate of that absorption can be calculated in classical physics; the respective oscillator strength is assigned to be 1. The probability to absorb light by a quantum oscillator is scaled by this factor to result in 'oscillator strength'

Technically it is a case of energy dissipation by a damped oscillator...

$$\ddot{x} + 2\beta \dot{x} + \omega_o^2 x = \frac{eE_{loc}}{m_e} \cos(\omega t)$$

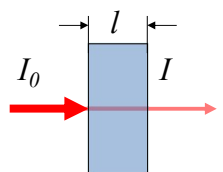
See: [https://www.cambridge.org/nl/files/6813/6689/9167/1076\\_161467.pdf](https://www.cambridge.org/nl/files/6813/6689/9167/1076_161467.pdf)

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## Oscillator strength vs extinction

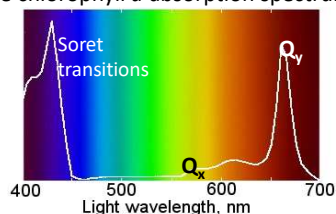
**Oscillator strength:**  $f = \frac{2}{3} \frac{m_e}{\hbar^2 e^2} \Delta E |\vec{\mu}|^2$        $\vec{\mu} \equiv e \langle \psi_b | \vec{r} | \psi_a \rangle$

How does it relate to the measured absorption spectrum?



$$A = \log \frac{I_0}{I} = \epsilon c l$$

The chlorophyll *a* absorption spectrum



**Oscillator strength:**  $f = 4.32 \cdot 10^{-9} \left[ \frac{\text{mol} \cdot \text{cm}^2}{L} \right] \int \epsilon(\nu) d\nu$

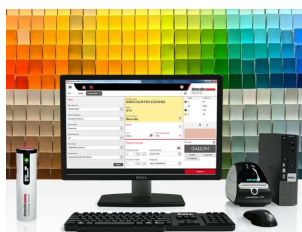
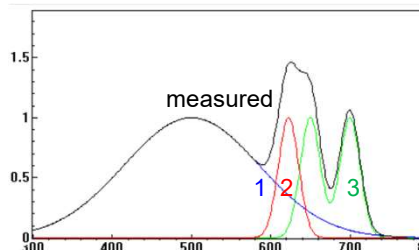
Note: spectral broadening does not change the total absorption of the integrated spectrum

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## Absorption of mixtures

Absorption of a mixture of noninteracting molecules is the sum of their absorbances. Knowing the spectra of constituents, the concentration of each can be deduced from that sum by solving the system of equations:

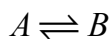
$$\begin{aligned}\epsilon_{1(\lambda 1)}C_1 + \epsilon_{2(\lambda 1)}C_2 + \epsilon_{3(\lambda 1)}C_3 + \dots &= A_{\lambda 1}/l \\ \epsilon_{1(\lambda 2)}C_1 + \epsilon_{2(\lambda 2)}C_2 + \epsilon_{3(\lambda 2)}C_3 + \dots &= A_{\lambda 2}/l \\ \epsilon_{1(\lambda 3)}C_1 + \epsilon_{2(\lambda 3)}C_2 + \epsilon_{3(\lambda 3)}C_3 + \dots &= A_{\lambda 3}/l\end{aligned}$$



Color matching in a paint store:  
Uses a spectrophotometer to measure the reference spectrum and fits the spectrum using different paints (strictly speaking it looks at a reflectance spectrum)

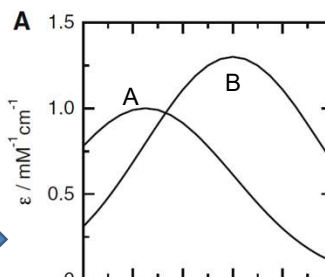
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## Absorption of mixtures

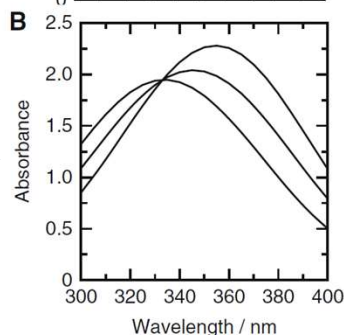


Conversion of species A into B: an isosbestic point is a wavelength  $\lambda_i$  at which extinction  $\epsilon_A(\lambda_i) = \epsilon_B(\lambda_i)$

Pure substances A and B



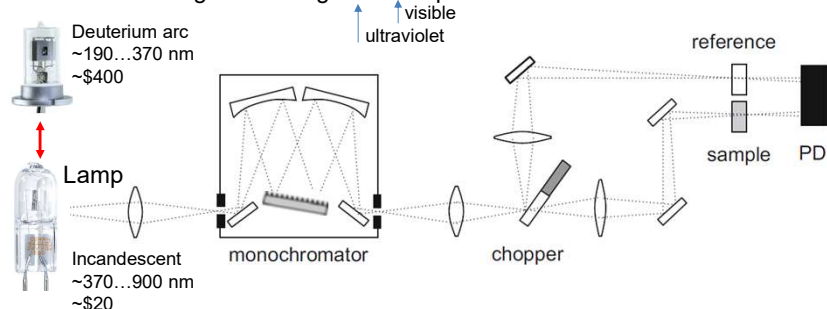
mixtures of A and B components in molar ratios of 1:3, 1:1 and 3:1  
Total concentration is kept constant  
 $C_A + C_B = \text{const}$



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## Steady state absorption spectrometer

Most common design: scanning "UV-Vis spectrometer"



Beckman DU640 UV/Vis

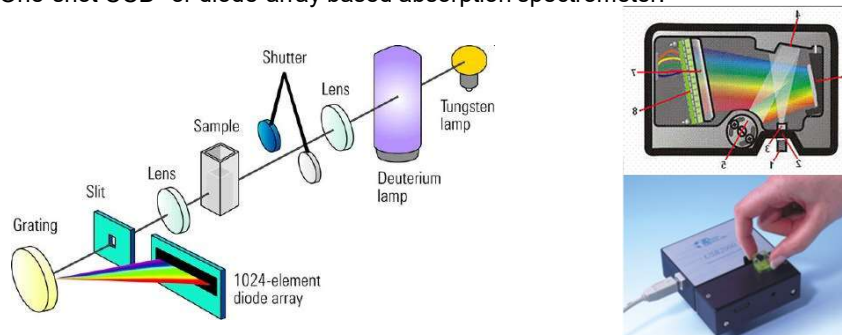
Major specs:

- Spectral range. Typical 190-900 nm
- Spectral resolution: 1-10 nm
- Dynamic range in A: 3-5 orders

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## Steady state absorption spectrometer

One-shot CCD- or diode-array based absorption spectrometer:



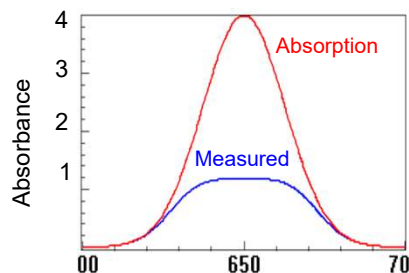
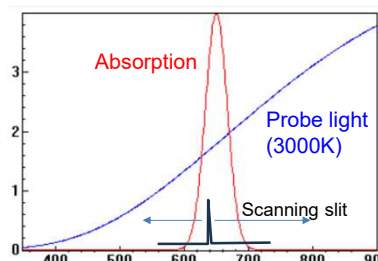
Ocean Optics USB2000 UV - Vis

A common problem of UV-VIS spectrometers:

detectors in IR range  $>1.1\mu$  are not sensitive, cannot be used for infrared range

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## UV-VIS spectrometer: saturation



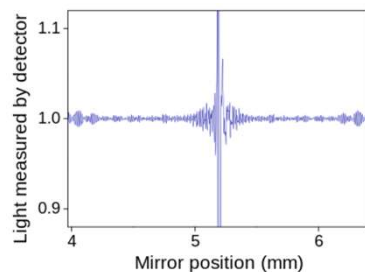
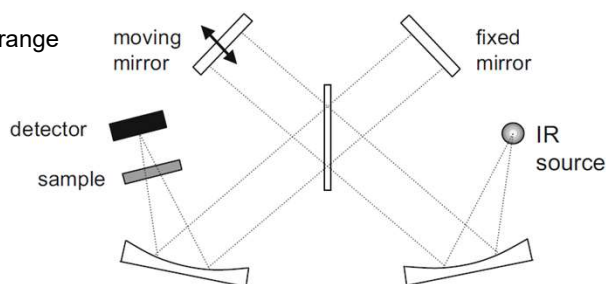
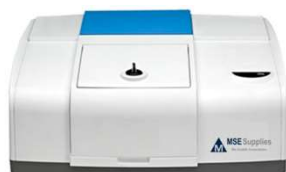
Contrast: the ability of a spectrometer to block light out of its bandpass range.

Even if the dynamic range is specified to be 5 (max  $A = 5$ ), some of the broad probe light can leak through and be detected by PMT as a background. High absorption range will be distorted.

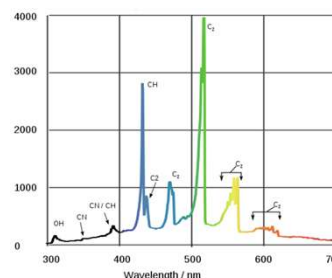
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## Fourier-transform IR spectrometer

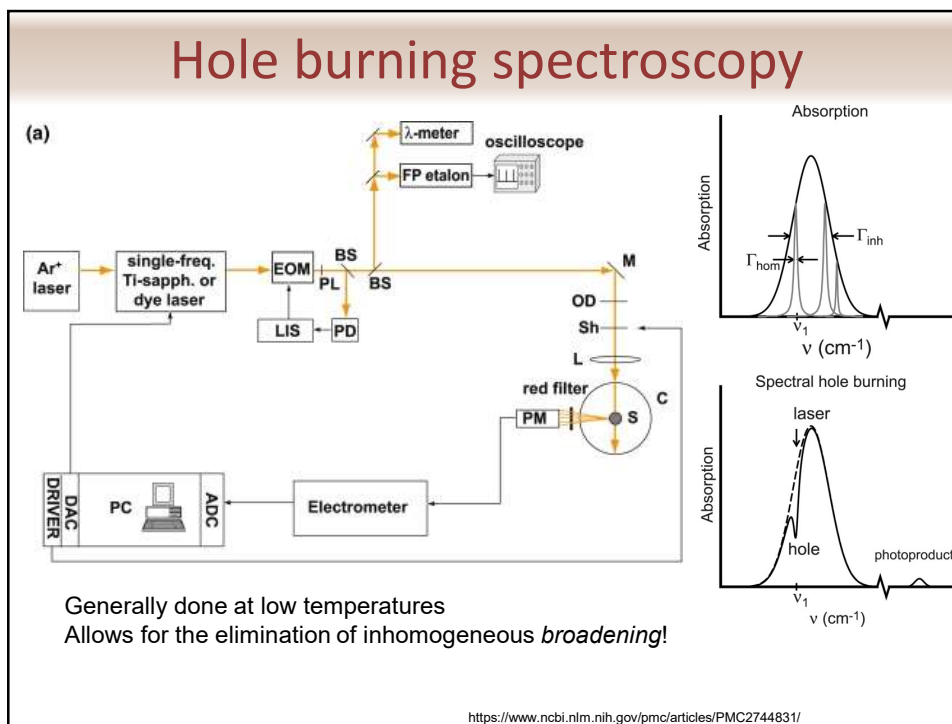
FTIR spectrometer:  
works well in the InfraRed range



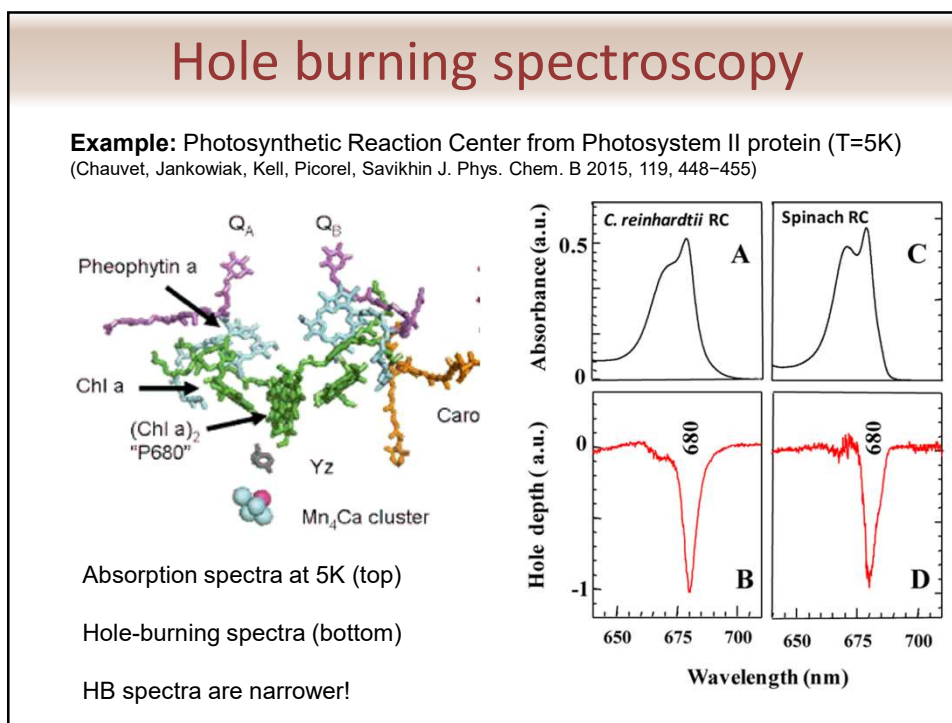
Fourier  
transform



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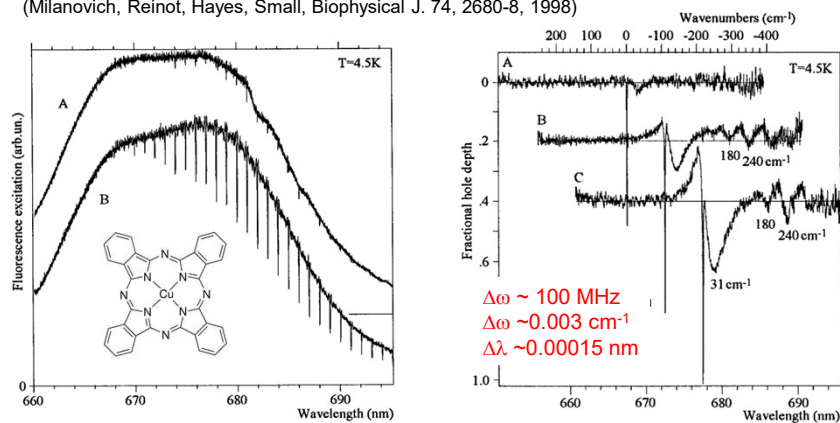


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## Hole burning spectroscopy

**Example:** Aluminum Phthalocyanine Tetrasulfonate in MCF-10F, Human Breast Epithelial Cells: A Hole Burning Study.

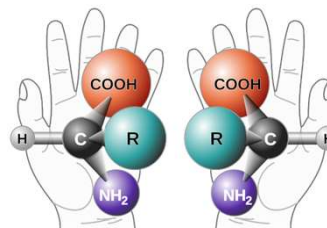
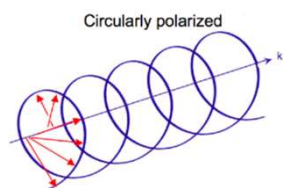
(Milanovich, Reinot, Hayes, Small, Biophysical J. 74, 2680-8, 1998)



Action spectra of holes burned across the fluorescence excitation profile of APT in MCF-10F cells

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## Circular Dichroism spectroscopy



Two enantiomer amino acids: same chemical structure, but different handedness, L- and D- amino acids

*D* = dexter (latin) – right

*L* = laevus (latin) left

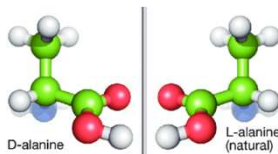
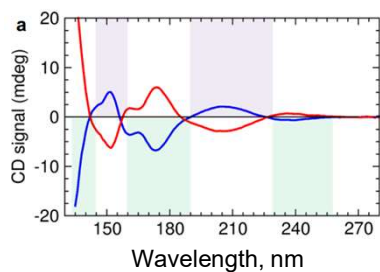
Most amino acids in living organisms are L-

CD spectrometer measures the difference in absorption of right and left circularly polarized light. The difference in absorption is typically 2-3 orders smaller than conventional absorption.

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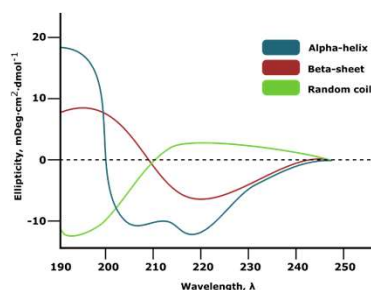


## Circular Dichroism spectroscopy



CD spectra of D-alanine (red); L-alanine (blue)

<https://www.nature.com/articles/s41467-022-28184-0>



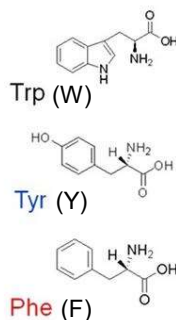
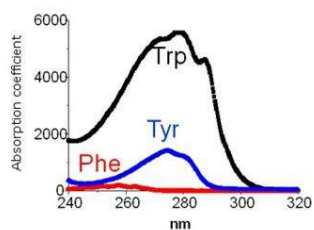
CD as a protein secondary structure probe

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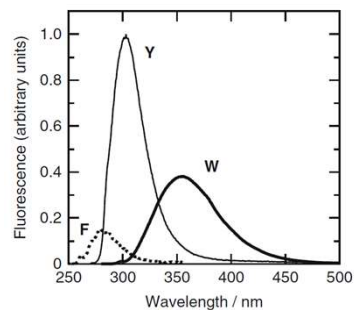
## Protein absorption and emission: UV

Example:

Absorbance



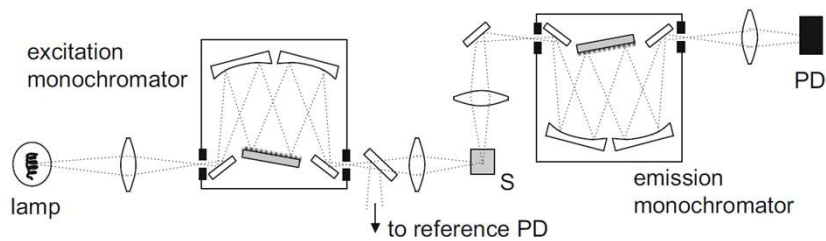
Emission



[https://regi.tankonyvtar.hu/hu/tartalom/tamop412A/2011-0073\\_introduction\\_practical\\_biochemistry/ch04s06.html](https://regi.tankonyvtar.hu/hu/tartalom/tamop412A/2011-0073_introduction_practical_biochemistry/ch04s06.html)

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## Fluorimeter: emission spectrum

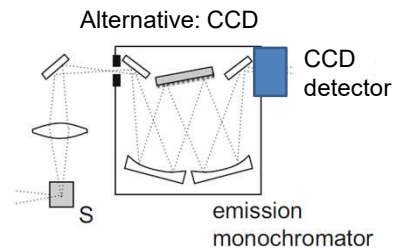


### Detectors:

Photodiode: low sensitivity, noise  $>10^4$  photons  
high dynamic range

Photomultiplier: high sensitivity, noise  $<10^2$  photons  
low dynamic range

CCD- charge-coupled device  
high sensitivity 5-1000 photons  
low dynamic range  
array: can have thousands of pixels



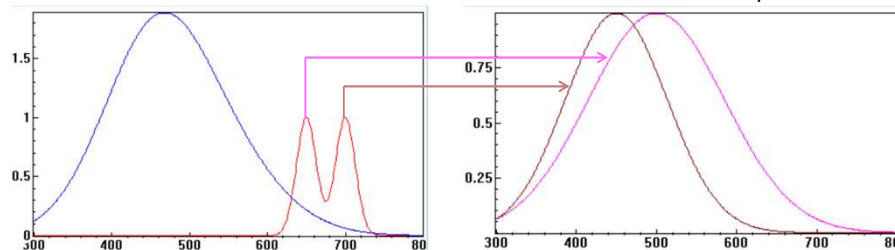
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## Emission excitation spectra

Here one scans the *excitation* wavelength while recording an emission signal at a fixed wavelength

Example:

Red – emission, blue - absorption



Question: do the 2 bands in emission come from the same species, or there is a mix of two?

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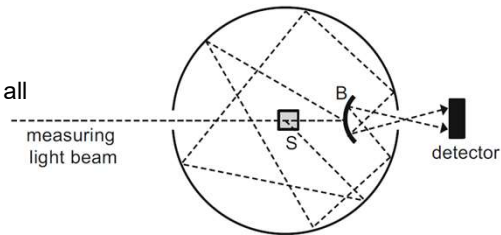
## Biological samples: distortion of absorption

### Light Scattering:

Cells can be large, if comparable to wavelength of light → scattering

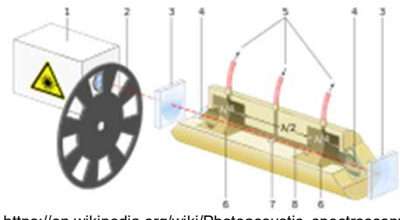
#### Some ways to minimize:

1. Can use *integrating sphere*: collect all light even if it is scattered



#### 2. Photoacoustic spectroscopy:

When sample is excited, heat is dissipated and immediate surrounding of the molecule *expands* causing acoustic wave. This can be detected by a microphone.



[https://en.wikipedia.org/wiki/Photoacoustic\\_spectroscopy](https://en.wikipedia.org/wiki/Photoacoustic_spectroscopy)

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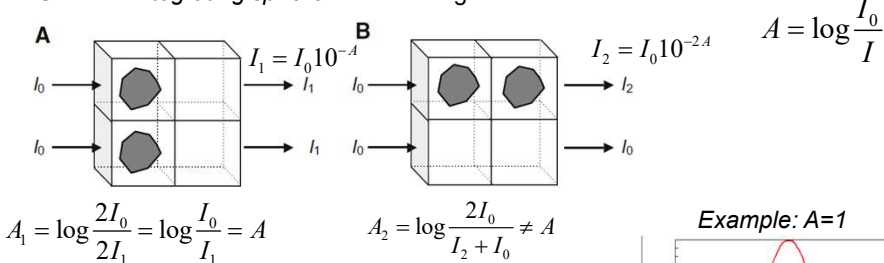
## Biological samples: distortion of absorption

### Nonuniform concentration:

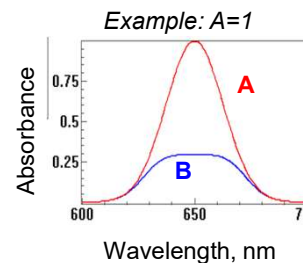
Molecules of interest may be clustered in distinct areas (like cells)

#### Some ways to minimize:

1. Can use *integrating sphere*: collect all light even if it is scattered



The effect: absorption gets “saturated” at maxima.  
Band tops get “flattened”



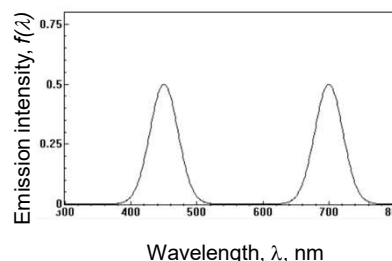
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## Emission spectrum: signal & scales

Most spectrometers measure emission intensity in a *wavelength* scale.

$f(\lambda)d\lambda$  = emission intensity integrated over  $d\lambda$  around  $\lambda$ .

Unlike the absorption spectrum, the emission spectral shape depends on UNITS!!!



**Emission intensity (y-scale):** can be measured as the *power* of emitted radiation, or as the number of photons emitted per unit time

- Transformation between Y-units:  $f(\text{power}) = f(\text{photons}) \times \hbar\omega$
- to convert the shape:  $f(\text{power}) \leftarrow f(\text{photons})/\lambda$

**X-scale:** can be wavelength (nm, Angstroms) or energy ( $\text{cm}^{-1}$ , eV)

$$\text{cm}^{-1} = 1/\text{cm} = 10^7/\text{nm}; \quad \text{eV} = 1239/\text{nm}$$

**Complications with X-scale** transformation: spectrum  $f(\lambda)d\lambda$  or  $f(\omega)d\omega$ . Since spectra are usually measured with a constant bandwidth  $d\lambda$ , the respective bandwidth in  $d\omega$  will become larger at shorter wavelengths!

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## Emission spectrum: signal & scales

Transformation of x-scale:  $f(\lambda)d\lambda$  to  $f(\omega)d\omega$

$$\omega = 2\pi\nu = \frac{2\pi c}{\lambda}$$

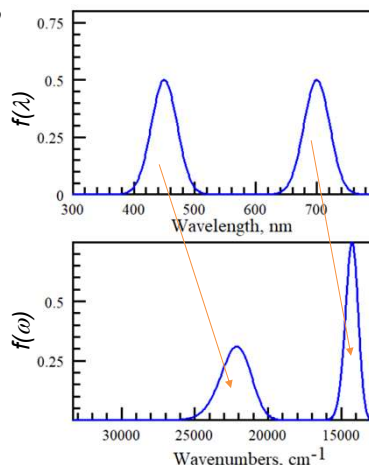
$$d\omega = (-)\frac{2\pi c}{\lambda^2} d\lambda$$

$$d\omega \propto \frac{d\lambda}{\lambda^2}$$

$$f(\lambda)d\lambda \propto f(\lambda)\lambda^2 d\omega$$

$$\downarrow$$

$$f(\omega) \propto f(\lambda)\lambda^2$$



When converting spectral shape from wavelength scale to energy scale:  
multiply spectrum by  $\lambda^2$  (or divide by  $\omega^2$ )

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## Emission spectrum: calibration

The spectral sensitivity of the emission spectrometer is typically calibrated using an incandescent light source at a known temperature - black-body radiation. A cheap way: get an incandescent light bulb with known temperature in Menards, the best would be 12-24V lamp used in projectors.

NOTE: the Planck curve MUST be modeled in the correct units used by the spectrometer!!!

This Planck equation is in emission energy per unit interval of photon energy!

$$\rho(\omega) = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\hbar\omega/k_B T} - 1}$$

The equation for spontaneous emission is also in energy units!

$$A = \frac{\omega_0^3 |\mu|^2}{3\pi\epsilon_0 \hbar c^3}$$

For comparison with an experiment: need to match bot X and Y scales!

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## Emission spectra

NOTES:

- The experimental spectra often omit the vertical scale units! But typically, emission is measured using PMT or silicon detectors, and the raw signal is proportional to the number of photons (counts/second)
- The theorists also often omit vertical scale units, but most tend to work in energy scale!

Luminescence – a general term for any kind of emission of an unheated sample

Fluorescence – short-lived emission from singlet excited states (~1 ns)

Phosphorescence – long-lived emission from triplet states (microseconds to seconds)

Thermoluminescence – emission of light when a sample is being heated

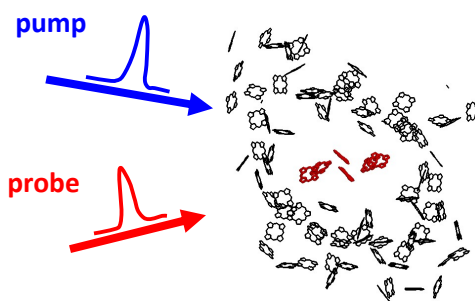
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## Time-resolved spectroscopy

- Pump-probe spectroscopy
- Time-correlated photon counting
- Fluorescence up-conversion

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## Pump-probe spectroscopy

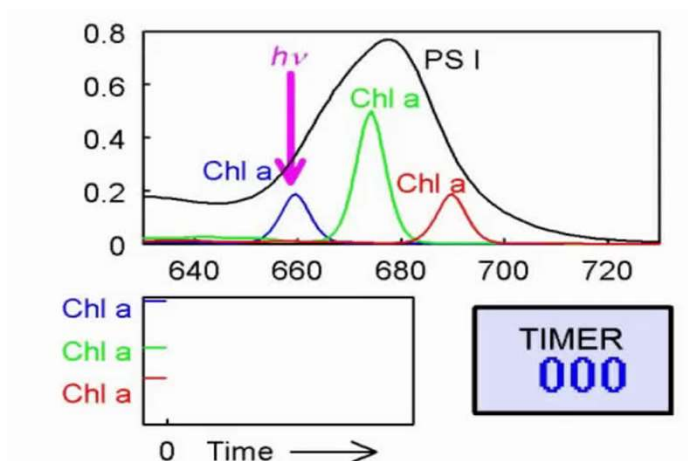


- **Pump pulse:** creates electronic excitation
- **Probe pulse:** probes the absorption changes due to excitation and electron transfer as a function of wavelength and time delay.

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## Pump-probe spectroscopy

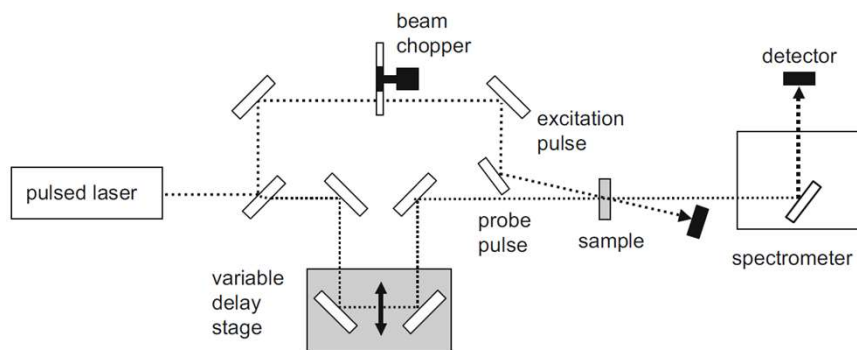
Energy transfer/decay is reflected in absorption change dynamics



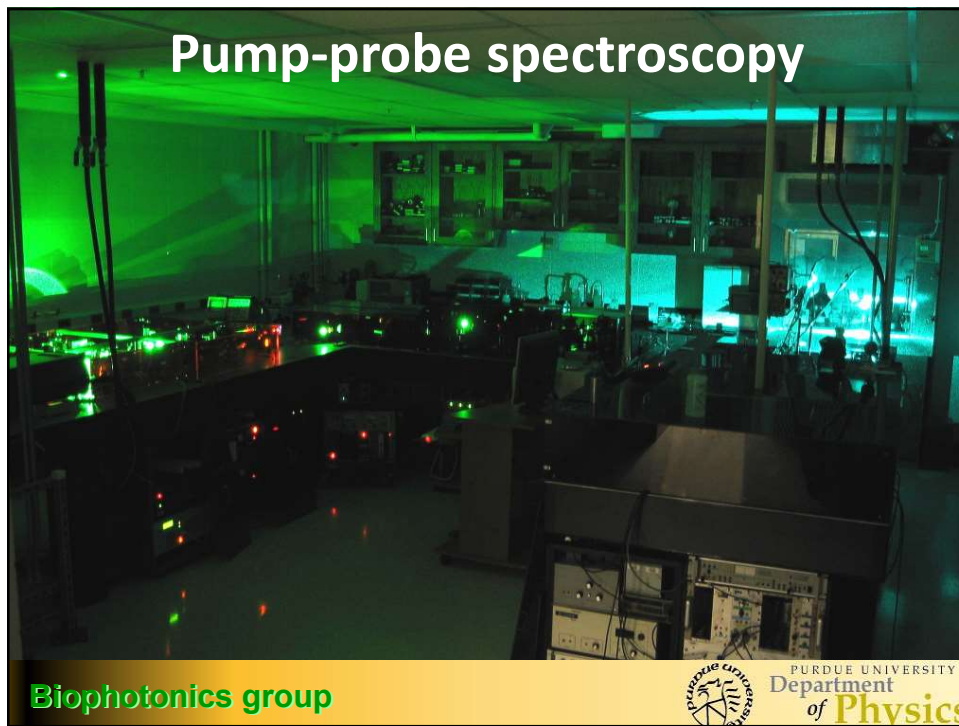
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## Pump-probe spectroscopy

Time resolution is defined by pulse length



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## Pump-probe spectroscopy

Home-built laser system provides light pulses that are shorter than 100 fs ( $<10^{-13}$  s)

How fast is 100 fs?

- Record a 1-second long movie with 100 fs/frame
- Will need 10,000 years to watch it through using regular VCR speed (at 30 fps)

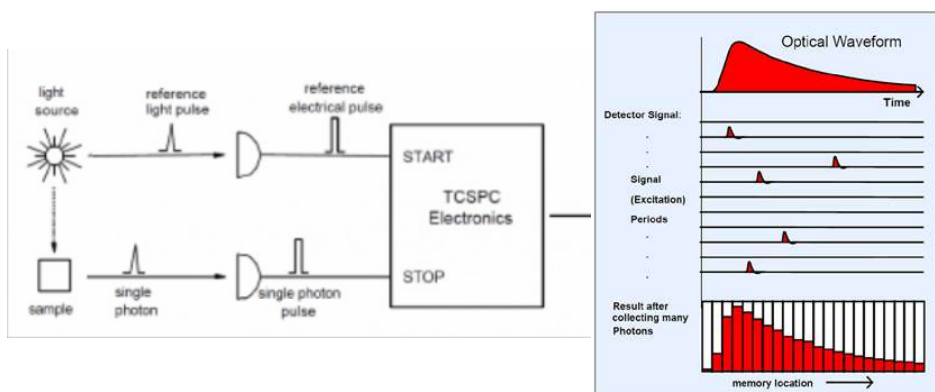


*Using 100 fs pulses we can see the motion of individual atoms*

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## Time-correlated photon counting

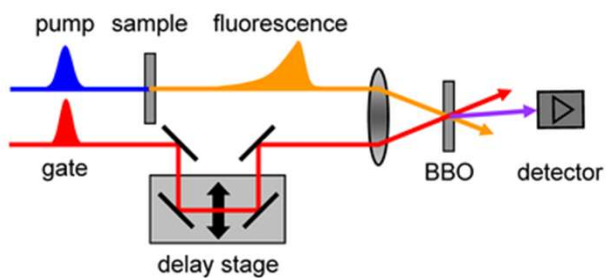


Time resolution down to  $\sim 10$  ps

<https://www.edinst.com/us/blog/what-is-tcspc/>

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## Fluorescence up-conversion



Time resolution can be  $< 100$  fs

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## Homework 1

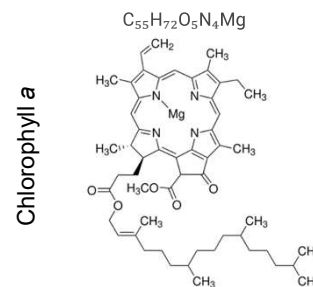
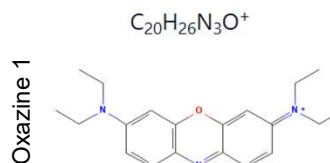
*Determine the transition dipole strength of a molecule experimentally and computationally*

### Homework 1: Experimental

Subject: laser dye oxazine 1 (or 725)  
3,7-bis(diethylamino)phenoxazin-5-ium perchlorate  
MW 423.89

Why oxazine?

- cheaper (~\$9/1mg vs \$220/1mg for Chl a)
- not too large for computation
- Contains typical motifs of organic molecules found in life

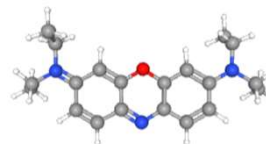


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## Homework 1

### Homework 1: Experimental

Subject: laser dye Oxazine 1 (or 725)  
Task: Determine the transition dipole strength of a molecule



1. Prepare a sample dissolved in MeOH (methanol) of known concentration
  - Use a scale to weigh the sample (powder)
  - Determine # of moles in the sample
  - Dilute in MeOH of known volume
  - Calculate concentration
2. Measure its absorption spectrum using a standard absorption spectrometer.
3. Convert spectra to appropriate units (extinction coefficient vs wavenumbers)
4. Use equations introduced in Lecture 4 to calculate oscillator strength and transition dipole moment

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